Automatic Tagging of Word Senses for a Large-Scale Historical Japanese Corpus

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A large-scale historical corpus annotated with word sense tags is invaluable for the study of classical texts, especially when morphological analysis is incomplete or unreliable. This is essential for historical language research, where no native speakers can confirm intended meanings. However, manual word sense annotation is particularly challenging in classical languages, requiring expert interpretation of nuanced context. To address this, we developed a system that automatically assigns word sense tags to every content word in a 20-million-word corpus of historical Japanese. Our method uses a Bidirectional Encoder Representations from Transformers (BERT) based model (Devlin et al. 2019) and achieves an accuracy of 88.57%. The system operates on tokenized input and directly labels sub-word tokens, avoiding reliance on detailed morphological annotation—making it suitable for diachronic corpus construction where annotation standards may vary over time..

Data

The corpus used in this study is the Corpus of Historical Japanese (CHJ) (NINJAL, Japan. 2022) . Portions of CHJ have been manually annotated with word senses, known as CHJ-WLSP (Asahara et al. 2022) . For word sense tags, we utilized concept IDs provided by the Word List by Semantic Principles (WLSP) (NINJAL, Japan. 2004) , a comprehensive Japanese thesaurus containing 98,214 entries categorized by syntactic and hierarchical semantic principles. concept ID is represented by a five-digit number. For example, concept ID 1.1920 represents “Nominal - relation - quantity - degree/limit.“ This concept ID is used as a word sense tag. Additionally, the historical version of the WLSP, known as the Nihon Koten Taisho Bunrui Goihyo (hWLSP) (Miyajima et al. 2014), includes a thesaurus based on historical word senses, aligned with the vocabulary of the period. Together, these two resources integrate contemporary and historical vocabulary under a unified word sense hierarchy.

Table 1 provides an overview of CHJ’s statistics, while Table 2 presents details of CHJ-WLSP (Asada et al. 2023). The CHJ corpus consists of 110 document files, and CHJ-WLSP comprises ten specific works, listed in Table 3 (Asahara et al. 2022) .

Table 1. Statistics of CHJ

Tokens	20,771,468
Types	135,705
Target tokens	14,918,386
Target types	5,785

Table 2. Statistics of CHJ-WLSP

Tokens	647,751
Target tokens	329,109
Target types	3,878
Target sense types	1,747
Average of senses	2.91

Table 3. The Name, Year, and Number of the Literature Book Used for the Experiments

Name	Year	Number of Words
Taketori Monogatari(lit. The Tale of the Bamboo Cutter)	900	12,757
Tosa Nikki (lit.Tosa Diary)	934	8,208
KonjakuMonogatari-shu(lit. Anthology of Tales from the Past)	1100	175,598
Hojoki (lit. Square-jo Record)	1212	5,402
Uji Shui Monogatari (lit. Gleanings from Uji Dainagon Monogatari)	1220	120,705
Jikkin-sho(AMiscellanyofTenMaxims)	1252	90,177
Tsurezuregusa (Essays in Idleness)	1336	40,834
Toraakira-bon-Kyogen	1642	5,448
TaiyoTheSun (Magazine)	1895	6,394
1st Jinjo Shogaku Tokuhon (Textbook)	1904	45,334
2nd Jinjo Shogaku Tokuhon (Textbook)	1910	96,894
Total		647,751

To evaluate the performance of our word sense disambiguation system, we used CHJ-WLSP as both training and validation data, with the Corpus of Historical Japanese (CHJ) as our target corpus.

Additionally, we incorporated BCCWJ-WLSP (Maekawa et al. 2014) for supplementary training data. BCCWJ-WLSP is a sense-tagged corpus derived from the Balanced Corpus of Contemporary Japanese (BCCWJ) [8], which provides a comprehensive resource of contemporary Japanese usage. Table 4 presents a summary of the statistics for CHJ-WLSP, as derived and calculated from (Maekawa et al. 2014).

Table 4. Statistics of BCCWJ-WLSP

Tokens	182,166
Types	20,010
Target tokens	77,344
Target types	4,573
Average of senses	3.24

Evaluation

We developed two models: one trained exclusively on CHJ-WLSP and the other using both CHJ-WLSP and BCCWJ-WLSP as training data. For validation, we used 1/5 of CHJ-WLSP, with the remaining portion dedicated to training. The approach adopted here follows that proposed in (Asada et al. 2023), employing a sequential labeling method that incorporates document IDs. We believe that document information helps reduce the impact of data diversity. While the entire CHJ corpus serves as our target, we evaluated the model using CHJ-WLSP for validation, as it represents the annotated segment of CHJ. Notably, the definitions of morphological information between CHJ and CHJ-WLSP differ, as morphological definitions have evolved post-annotation of CHJ-WLSP. However, BERT-based models operate without detailed morphological information, allowing accurate prediction of word sense tags. Since both CHJ and BCCWJ were manually tokenized, tokenized sentences were used as input; the BERT tokenizer then segmented them further, with word senses assigned to each sub-word unit. We used https://huggingface.co/cl-tohoku/bert-base-japanese-v2 . The results trained with BERT were better than those trained with RoBERTa.

Table 5 presents the accuracy scores on the validation set, where a model trained on both CHJ-WLSP and BCCWJ-WLSP achieved an accuracy of 88.57%. For both models, the optimal learning rate was 1e-05, with an epoch count of 15. Accuracy was calculated as the proportion of correctly tagged word senses out of the total number of polysemous words. Notably, the number of target polysemous words increased when BCCWJ-WLSP was included in the training data alongside CHJ-WLSP, enhancing the model's generalization. Given our aim to annotate the full CHJ corpus, we selected the model trained with both corpora to accommodate potential out-of-vocabulary words from CHJ-WLSP present in the newer CHJ dataset.

Table 5. Accuracy on the Validation Set

Training Corpus	Accuracy	Correct	Target
CHJ -WLSP	85.25%	275,837	323,547
CHJ -WLSP +BCCWJ -WLSP	88.57%	430,496	486,049

Table 6 lists the three most frequently tagged word senses within CHJ. Concept IDs 2.3430 and 2.12000 correspond to verbs such as ## (suru, "do") and ##### (gozaimasu, "be"), which are also prevalent in BCCWJ. Conversely, Concept ID 1.2301, which represents terms for people, appears less frequently in BCCWJ. We plan to make CHJ accessible via a web-based search system that allows queries based on word meanings.

Table 6. The Three Most Frequently Tagged Word Senses within CHJ.

Rank	Concept ID	Description	Number
1	1.2301	Nominal-Subject-Human-Citizen	47,496
2	2.3430	Verbal-Action-Activity-Activity/Action	21,619
3	2.1200	Verbal-Relation-Existence-Existence	18,143

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